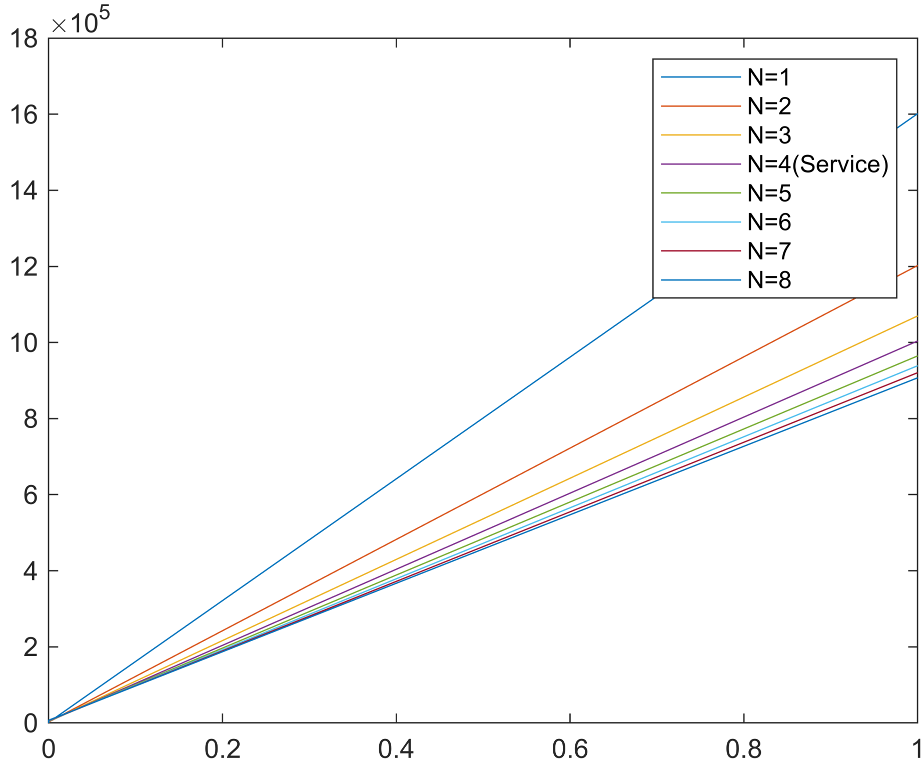
Exercise 1

- N=8 sensors each produce periodically 100 bits of measurements data every T=1mS.
- Data are to be sent across a shared communication line offering s= 1Mbps of serialization speed (bandwidth).
- Data from each sensor will be packetized with N (tbd) data points together with an overhead (header, error detection …) of O=100 bits
- Determine the best value for N taking into account delay and bandwidth.
- For arbitrary (variable) N, determine a DNC arrival constraint for each sensor as well as the aggregate arrival curve for all sensors.

Exercise 2

- Encode the arrival constraint of the aggregated flow (from ex 1) for all sensors as an rtccurve
- Encode the service curve of the communication line into rtc.
- Use rtch and rtcploth to determine the experienced delay
- Use rtcv and rtcplotv to determine the experienced backlog
- Use an appropriate tool in the rtc-toolbox to determine the optimal value for N

```
for N = 1:8
    r = 8*(100000 + 100000/N);
    burst = 8*(100*N + 100);

    alpha{N} = rtccurve([0,burst,r]);
end
rtcplot(alpha{:})
legend('N=1','N=2','N=3','N=4(Service)','N=5','N=6','N=7','N=8')
```



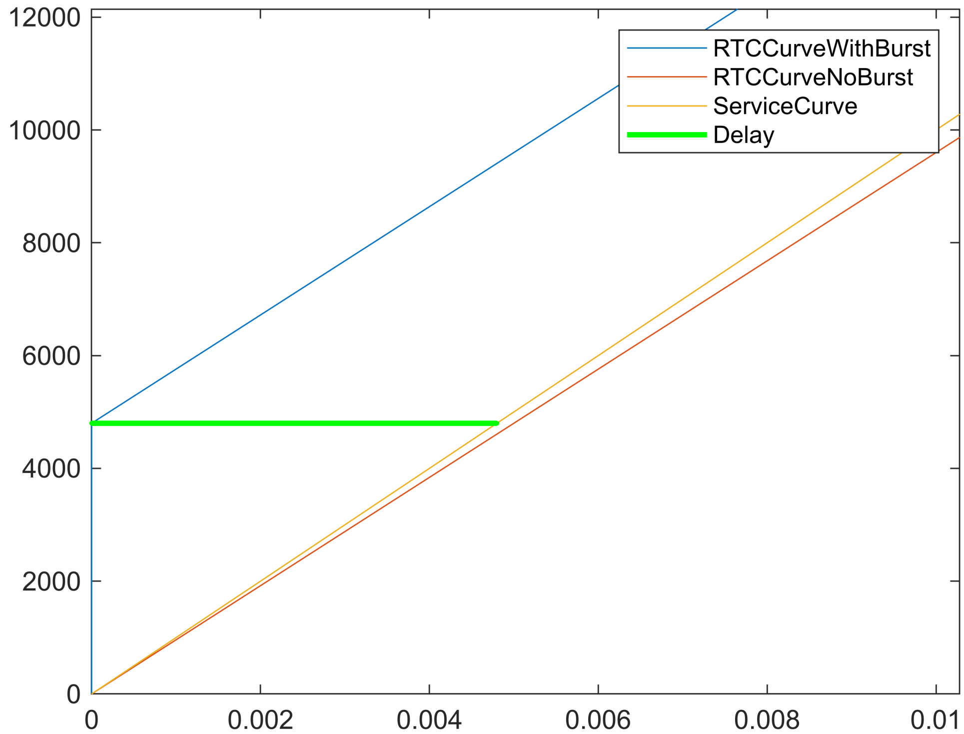

```
N = 5;

r = 8*(100000 + 100000/N);
b = 8*(100*N + 100)
```

```
b =
4800
```

```
a = rtccurve([0,b,r]);
noBurst = rtccurve([0,0,r]);
beta = rtccurve([0,0,1e6]); %ServiceCurve

rtcplot(a, noBurst, beta)


D = rtch(a,beta)
```

```
D =
0.0048
```

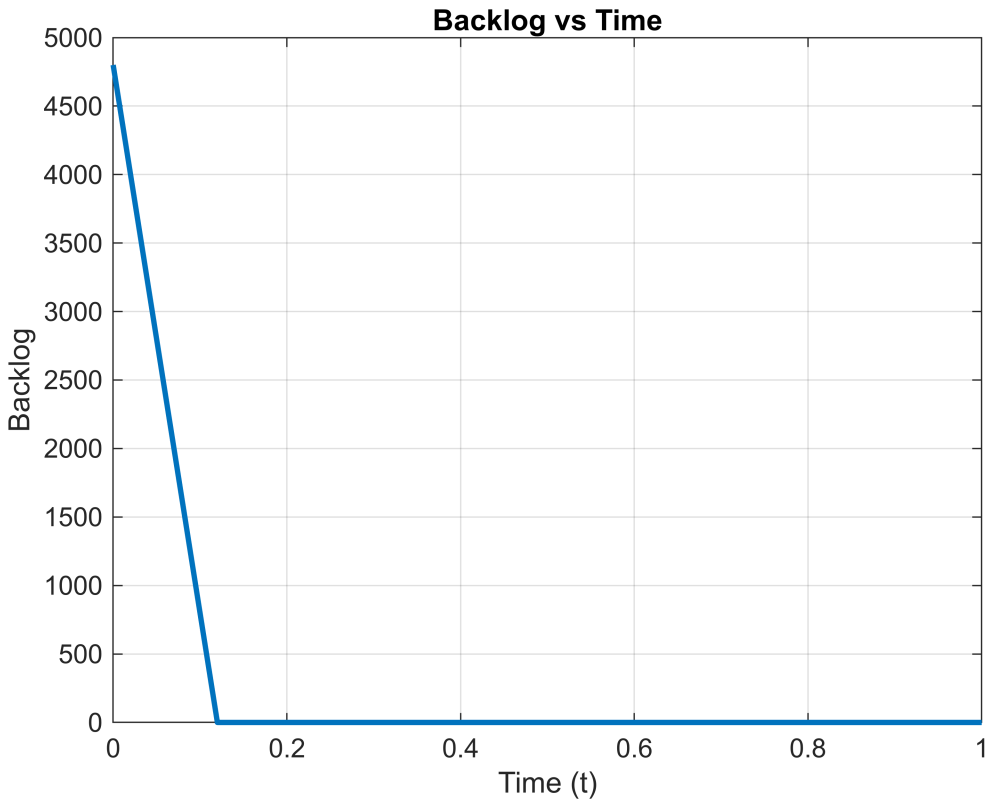
```
rtcploth(a,beta);

legend('RTCCurveWithBurst','RTCCurveNoBurst', 'ServiceCurve', 'Delay')
xlim([0.00000 0.01028])
ylim([0 12144])
```




```
B = rtcv(a,beta)
```

```
B =
4800
```

```
% Time vector
t = 0:0.001:1;  % t from 0 to 100

% Compute backlog
backlog = max(B + (r - 1e6) * t,0);

% Plot
plot(t, backlog, 'LineWidth', 2)
xlabel('Time (t)')
ylabel('Backlog')
title('Backlog vs Time')
grid on
```




```
beta = rtccurve([0,0,1e6]);

Nvals = 1:20;
delay = NaN(size(Nvals));

for N = Nvals

    r = 8*(100000 + 100000/N);
    b = 8*(100*N + 100);

    % skip infeasible solutions
    if r >= 1e6
        continue
    end

    alpha = rtccurve([0,b,r]);

    delay(N) = rtch(alpha,beta);

end

[minDelay,idx] = min(delay);

optimalN = Nvals(idx)
```

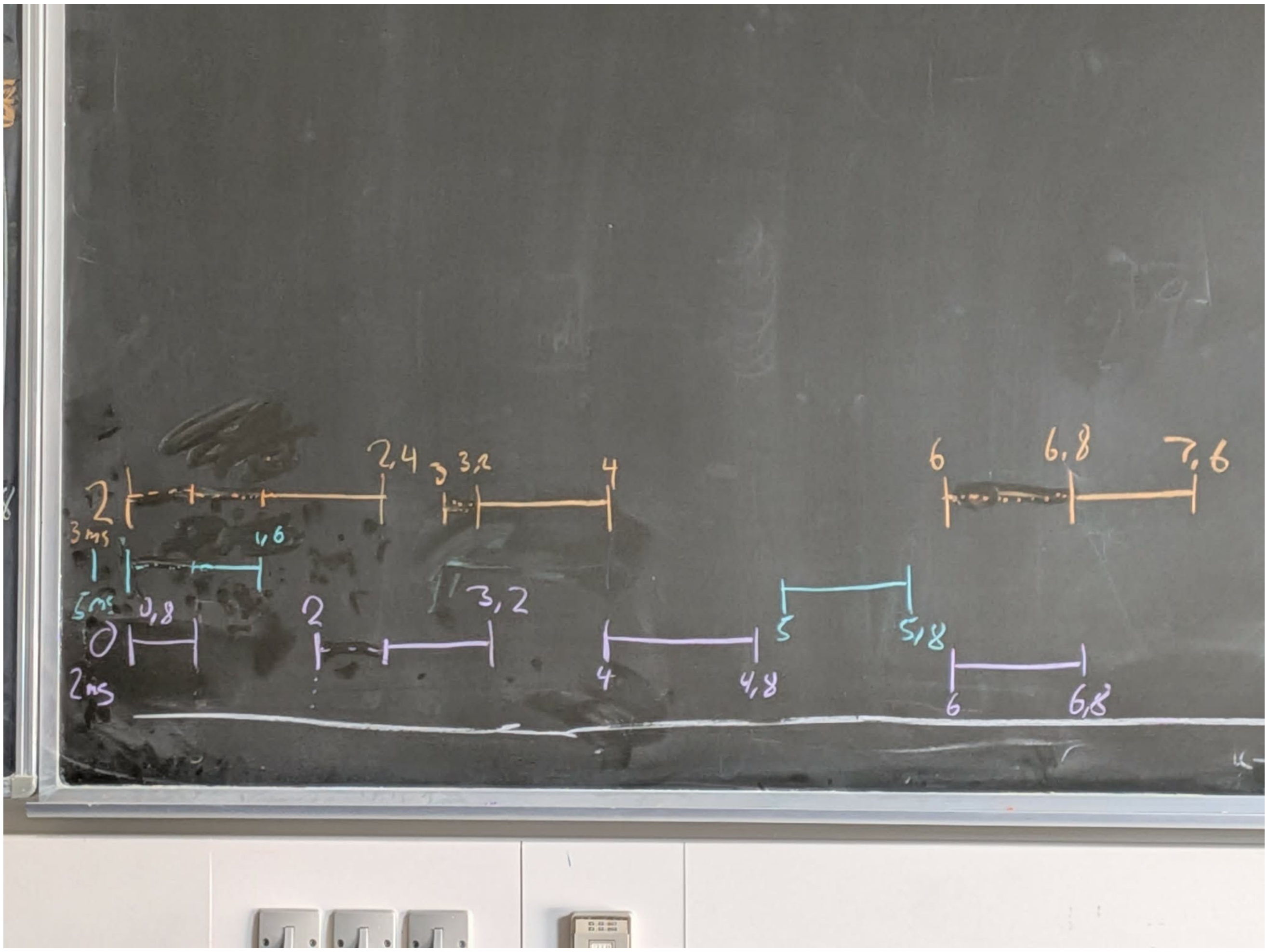
```
optimalN =
```

```
%Excercise 3
```

Exercise 3

- Consider 3 nodes on a single CANbus segment with the following periods and priorities:
  - 0: {2mS, 000...}
  - 2: {3mS, 0002...}
  - 1: {5mS, 0001...}
- Assume packet lengths of 100 bits.
- Determine the smallest bitrate on the CANbus for the traffic to be executable.
- Make an RTC-model for the 3 nodes and CANbus with arbitrary bitrate. Include a receive buffer in the lowest priority node as well as a periodic receiver with arbitrary period.
- Determine maximum p2p delay for node 2 as well as minimum receive buffer size (for not loosing packets) for arbitrary receive periods and bitrates.

```
%Packet processing time:
%T_P = P/Sr = 100 [b]/125 [kb/s] = 0.8 ms
```




```
%% Traffic parameters

L  = 100;       % bits

T0 = 2e-3;
T1 = 5e-3;
T2 = 3e-3;

%% Minimum average bitrate

Bmin = L*(1/T0 + 1/T1 + 1/T2);

fprintf('Minimum bitrate = %.2f bit/s\n',Bmin);
```

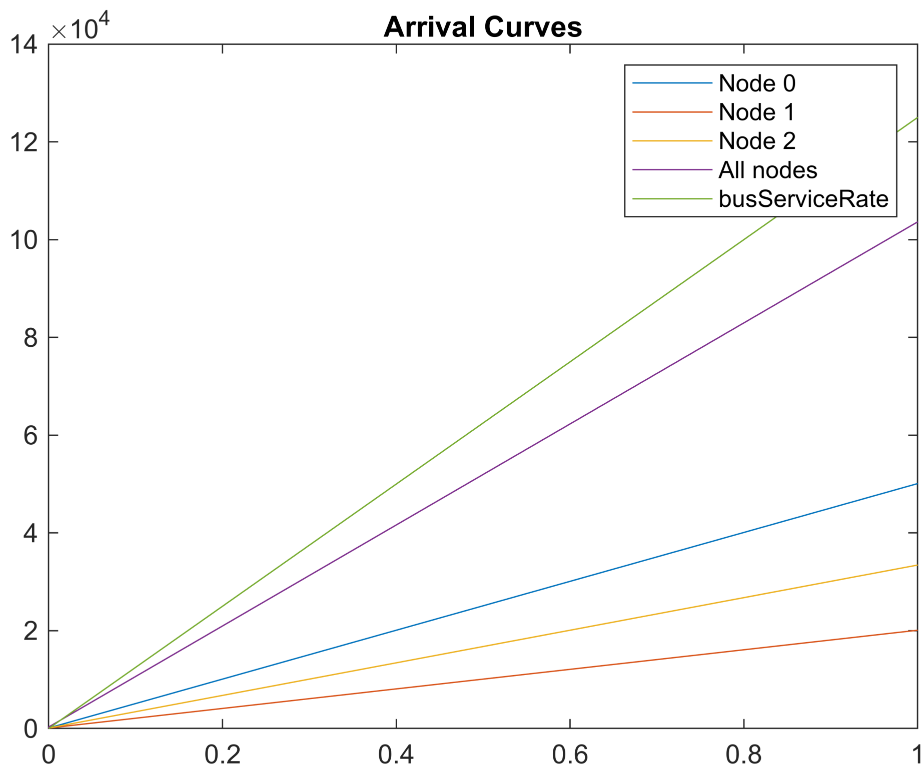
```
Minimum bitrate = 103333.33 bit/s
```

```
fprintf('Minimum bitrate = %.2f kbit/s\n',Bmin/1000);
```

```
Minimum bitrate = 103.33 kbit/s
```

```
%% Arrival curves

L = 100;
```

```
a0 = rtccurve([0,L,L/0.002]);   % highest priority
a1 = rtccurve([0,L,L/0.005]);   % medium priority
a2 = rtccurve([0,L,L/0.003]);   % lowest priority
a3 = rtccurve([0,3*L,Bmin]);

%% CAN bitrate

B = 125e3;

betaBus = rtccurve([0,0,B]);
figure(1)

rtcplot(a0,a1,a2,a3, betaBus)

title('Arrival Curves')
legend('Node 0','Node 1','Node 2','All nodes', 'busServiceRate')
```

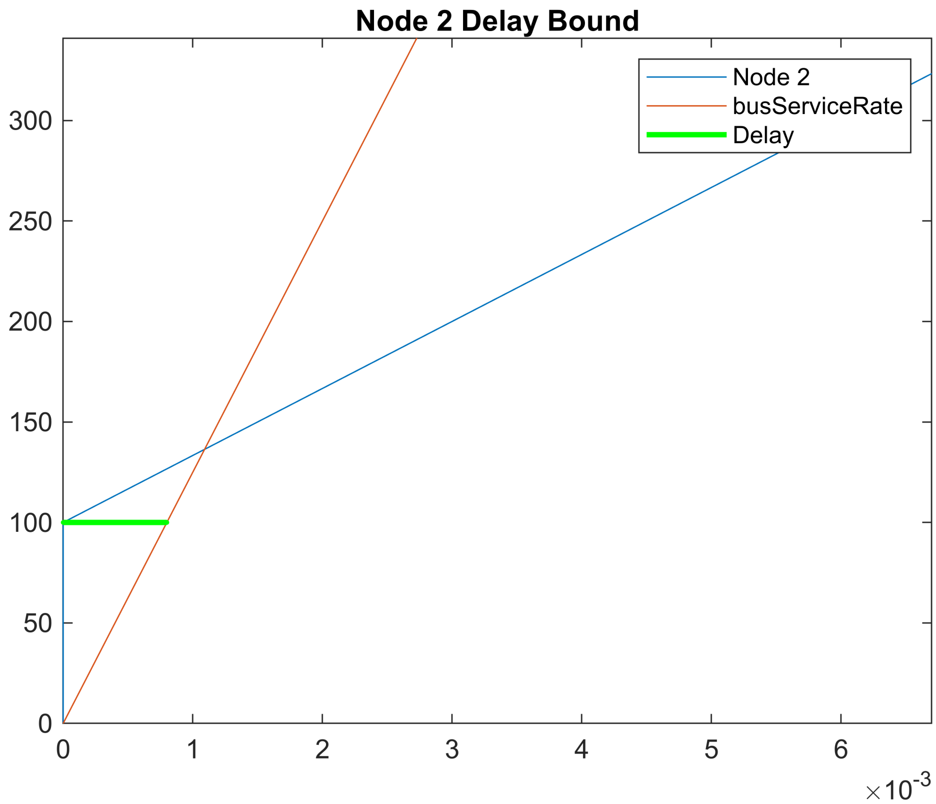



```
%% Delay bound

D2 = rtch(a2,betaBus);

fprintf('Node 2 bus delay = %.6f s\n',D2);
```

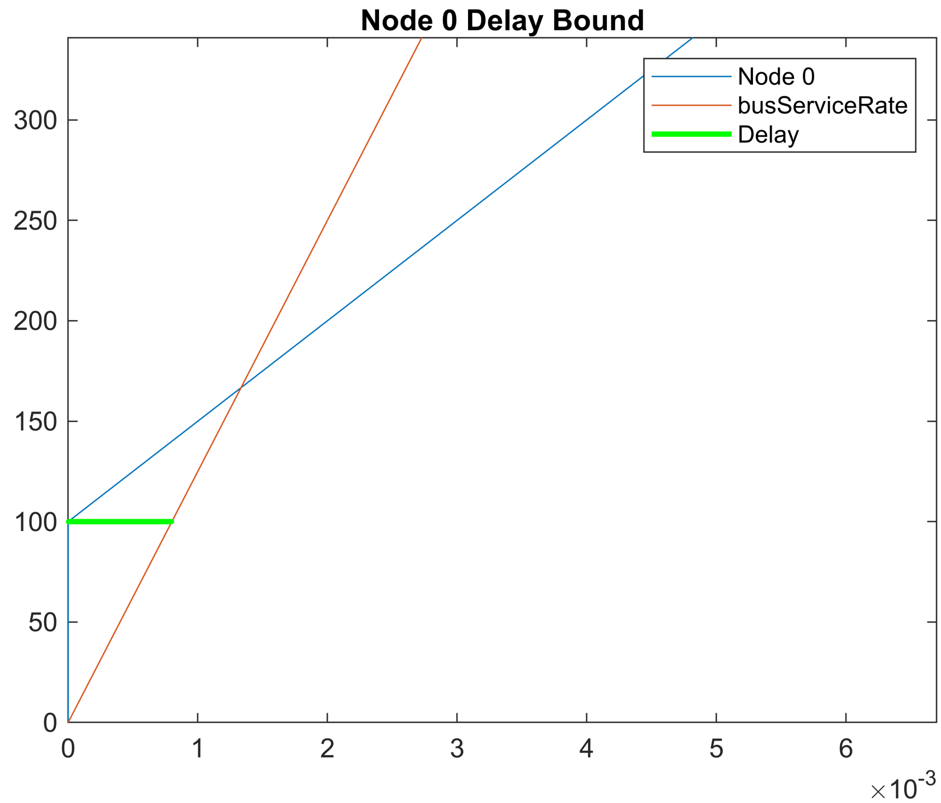
```
Node 2 bus delay = 0.000800 s
```

```
figure(3)
```

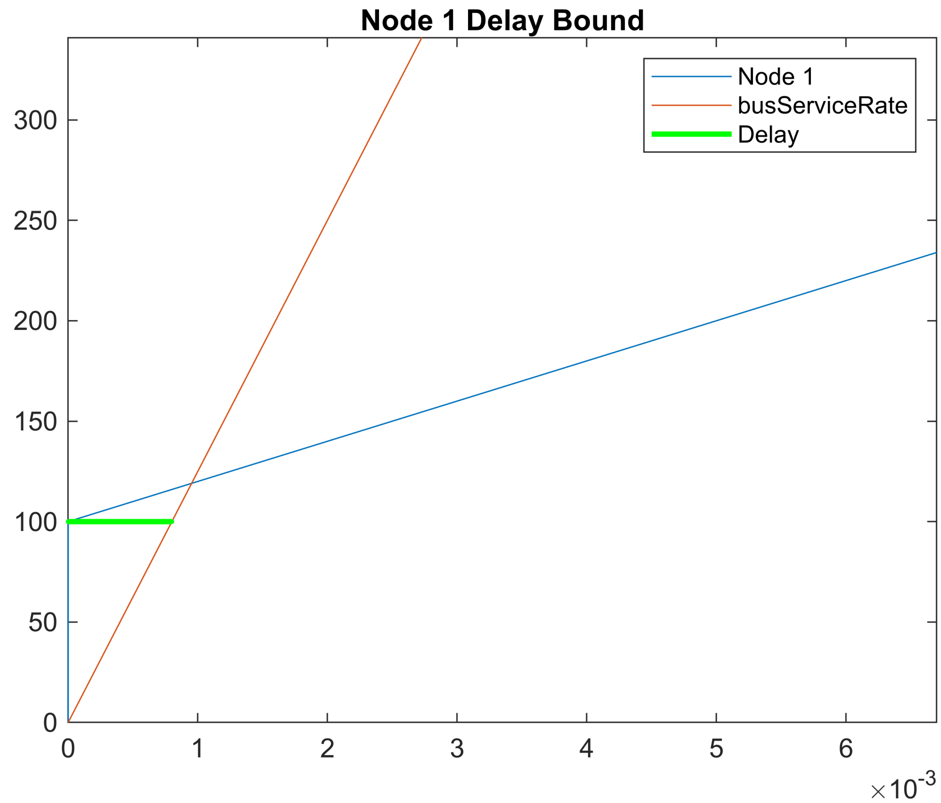
```
rtcplot(a2,betaBus)
rtcploth(a2,betaBus)

legend('Node 2', 'busServiceRate', 'Delay')
title('Node 2 Delay Bound')

xlim([0.00000 0.00670])
ylim([0 341])
zlim([-1.00 1.00])
```



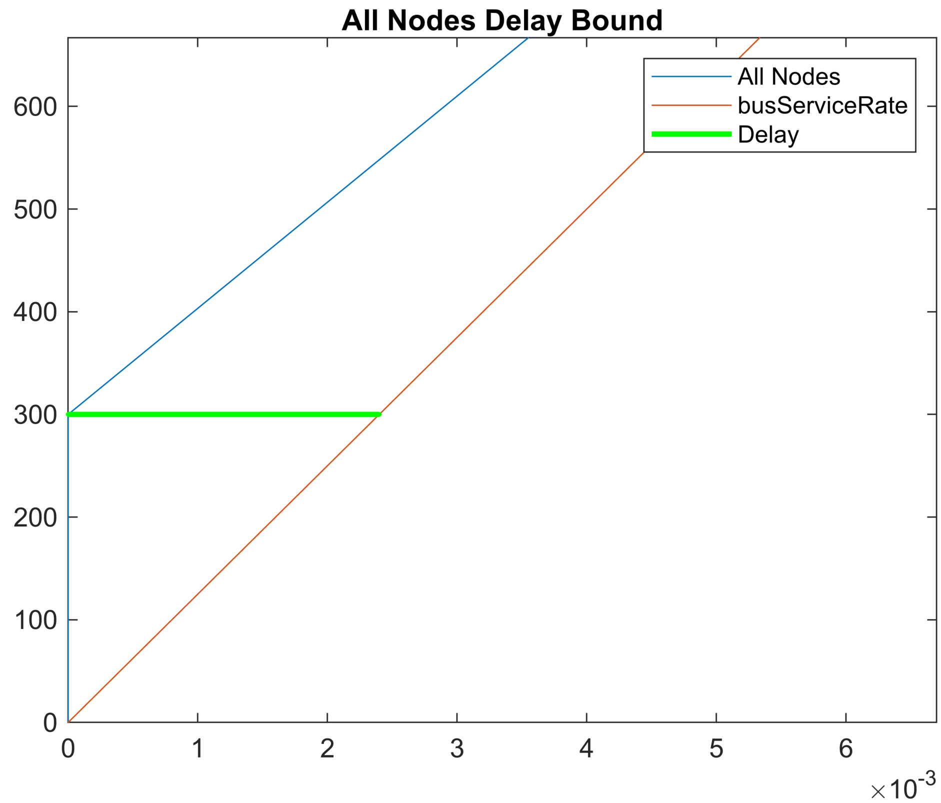

```
rtcplot(a0,betaBus)
rtcploth(a0,betaBus)

legend('Node 0', 'busServiceRate', 'Delay')
title('Node 0 Delay Bound')

xlim([0.00000 0.00670])
ylim([0 341])
zlim([-1.00 1.00])
```




```
rtcplot(a1,betaBus)
rtcploth(a1,betaBus)

legend('Node 1', 'busServiceRate', 'Delay')
title('Node 1 Delay Bound')

xlim([0.00000 0.00670])
ylim([0 341])
zlim([-1.00 1.00])
```




```
rtcplot(a3,betaBus)
rtcploth(a3,betaBus)

legend('All Nodes', 'busServiceRate', 'Delay')
title('All Nodes Delay Bound')

xlim([0.00000 0.00670])
ylim([0 341])
zlim([-1.00 1.00])

xlim([0.00000 0.00670])
ylim([0 667])
zlim([-1.00 1.00])
```




```
AvgRateN2 = L*(1/0.002+1/0.005)
```

```
AvgRateN2 =
70000
```

```
TB = 0.01;
maxBuffer = 2*L + TB*AvgRateN2
```

```
maxBuffer =
900
```